

TED(15)-6022

REVISION 2015

SIXTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING

HYDRAULIC MACHINES

MODEL QUESTION PAPER

(Maximum marks 100)

Time :3 hrs

PART-A

I Answer all questions in one or two sentences. Each question carries 2 marks

1. What is meant by efficiency of jet.
2. Name the components of pelton turbine
3. List the unit quantities of turbine

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2. Name the components of pelton turbine
3. List the unit quantities of turbine
4. Define specific speed of centrifugal pump.
5. Distinguish static head and dynamic head.

(5×2=10)

PART - B

(Maximum marks: 30)

II Answer any five of the following questions. Each question carries 6 marks.

1. Define the term impact of jets and jet propulsion
2. Derive an expression for work done by a jet of water impinging on a moving curved plate
3. Enumerate the advantages of water turbines
4. Explain the significance of specific speed of water turbines
5. Explain the main components of a reaction turbine
6. Discuss different types of casing of centrifugal pump
7. Describe the functions of air vessel

(5×6=30)

PART - C

(Maximum marks: 60)

(Answer one full question from each unit. Each full question carries 15 marks)

UNIT I

- III.a) Derive an expression for work done by a jet of water impinging on a moving flat vertical plate (7)
- b). A jet of water of 3.2 cm diameter moving with a velocity of 16 m/s strikes a hinged plate of weight 110 N at the centre of the plate. Find the angle through which the plate is swing (8)

OR

- IV a) Derive an expression for force exerted by the jet striking on stationary symmetrical curved plates at one end tangentially. (8)
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OR

IV a) Derive an expression for force exerted by the jet striking on stationary symmetrical curved plates at one end tangentially. (8)

b) A jet of water of diameter 50mm moving with a velocity of 20m/s strikes a fixed plate in such a way that the angle between the jet and plate is 70°. Find the force exerted by the jet on the plate (i) in the direction normal to the plate, and (ii) in the direction of jet. (7)

UNIT-II

V a) Explain the working of pelton wheel with sketch (7)

b). A Pelton wheel develops 2000 KW under effective head of 160 m and with an overall efficiency of 90%. Find the diameter of the nozzle, if coefficient of velocity for the nozzle is 0.95 (8)

OR

VI a). Discuss the different types of water turbines. (7)

b) .Explain the governing system used in impulse turbine (8)

UNIT-III

VII (a). Explain the working of Francis turbine with sketch (7)

(b). A turbine develops 2000 kilowatt under a head of 50m at 300 RPM. What is the specific speed? What would be its normal speed and output under a head of 20 m (8)

OR

VIII (a) Discribe the different types of draft tube (7)

(b) Compare impulse turbine and reaction (8)

UNIT IV

IX (a). Explain the working of air lift pump (7)

(b). A single acting reciprocating pump has a speed of 60 RPM. It has a plunger diameter of 200 mm and stroke of 300 mm. The suction and delivery heads are 5m and 16m respectively. Determine the actual power required to drive the pump if the efficiency of pump is 70 %. (8)

OR

X.(a) .Explain cavitation and priming of centrifugal pump (8)

(b) Describe the concept of NPSH (7)